

User Interface Design in the Waterfall Model

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Introduction

- The waterfall model follows a structured, step-by-step approach in software development.
- This presentation explores how UI design and development are handled at each stage of the SDLC.
- The key interface design/development activities, deliverables, and artifacts are described for each stage of the SDLC.

Planning Stage

- Goal: Define project scope, resources, and schedule.
- UI Designer Role: Provide input on UI feasibility and suggest tools or libraries.
- Key Activities: Stakeholder meetings, high-level UI goals.
- Artifacts/Deliverables: Project charter, initial UI considerations, risk assessments, feasibility study, team allocation document.

Requirements Definition Stage

- Goal: Capture the user's needs and define system functions.
- UI Designer Role: Conduct user research, interviews, and surveys.
- Key Activities: Storyboarding, user personas, use cases.
- Artifacts/Deliverables: Requirements documentation, UI use scenarios, functional and non-functional requirements, and user stories.

Design Stage

- Goal: Architect the system and design the user interface.
- UI Designer Role: Lead interface design—layout, navigation, interaction.
- Key Activities: Wireframes, prototypes, style guides.
- Artifacts/Deliverables: UI mockups, design documentation, high-fidelity prototypes, screen flow diagrams, user interaction models.

Implementation Stage

- Goal: Translate designs into working software.
- UI Designer Role: Work with developers to ensure UI accuracy.
- Key Activities: Front-end coding, asset creation, CSS/HTML/JS implementation.
- Artifacts/Deliverables: UI components, UI codebase, style sheets, assets library, developer documentation.

Integration & Testing

- Goal: Verify system functionality and usability.
- UI Designer Role: Test for usability, accessibility, and visual consistency.
- Key Activities: User testing, bug fixing, refinement.
- Artifacts/Deliverables: Usability reports, updated prototypes, accessibility compliance report, QA test logs, bug tracking documents.

Most Critical Stage

- The Design Stage is the most critical for UI because it defines the visual and interactive user experience that will set the tone for the rest of the development cycle.
- Decisions made here set the foundation for user satisfaction and usability.
- Strong design at this stage leads to fewer changes later and better user engagement.

Final Thoughts

- Each stage plays a unique role in UI design and development.
- Clear collaboration between designers, developers, and stakeholders throughout the SDLC ensures a smoother, more user-focused product.
- Designers must stay involved from planning to testing for the best results.

References

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